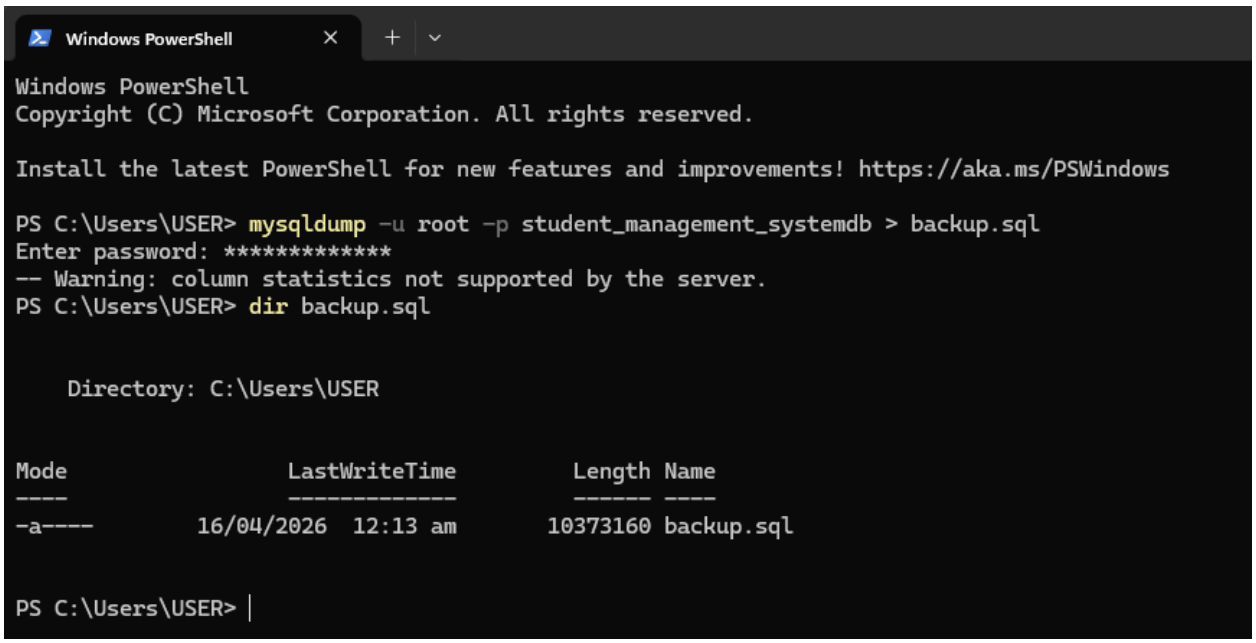
IT 105 – Information Management I

Phase 4: Cloud Backup and Integration of Emerging Database Technologies

1. **Backup**

```
mysqldump -u root -p student_management_systemdb > backup.sql
```



2. **Simulated Failure**

```
mysql -u root -p
DROP DATABASE student_management_systemdb;
```

```
Windows PowerShell

Windows PowerShell
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PS C:\Users\USER> mysql -u root -p
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 226
Server version: 5.5.5-10.4.32-MariaDB mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE student_management_systemdb;
Query OK, 11 rows affected (0.12 sec)

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| backup             |
| ibu-notice-system  |
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| studentsystemdb    |
| test               |
+--------------------+
8 rows in set (0.00 sec)

mysql> |
```

3. **Restore Database**

```
mysql -u root -p
CREATE DATABASE student_management_systemdb;
exit;
mysql -u root -p student_management_systemdb < backup.sql
```

```
C:\WINDOWS\system32\cmd.

C:\Users\USER>mysql -u root -p student_management_systemdb < backup.sql
Enter password: *************

C:\Users\USER>mysql -u root -p
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 234
Server version: 5.5.5-10.4.32-MariaDB mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+----------------------------+
| Database                   |
+----------------------------+
| backup                     |
| ibu-notice-system          |
| information_schema         |
| mysql                      |
| performance_schema         |
| phpmyadmin                 |
| student_management_systemdb |
| studentsystemdb            |
| test                       |
+----------------------------+
9 rows in set (0.00 sec)

mysql> |
```

4. Validation

```
SELECT
(SELECT COUNT(*) FROM users) +
(SELECT COUNT(*) FROM students) +
(SELECT COUNT(*) FROM instructors) +
(SELECT COUNT(*) FROM courses) +
(SELECT COUNT(*) FROM enrollments) +
(SELECT COUNT(*) FROM audit_logs)
AS total_records;
```

```
Windows PowerShell

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PS C:\Users\USER> mysql -u root -p
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 244
Server version: 5.5.5-10.4.32-MariaDB mariadb.org binary distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE student_management_systemdb;
Database changed
mysql> SELECT (SELECT COUNT(*) FROM users) + (SELECT COUNT(*) FROM students) + (SELECT COUNT(*) FROM instructors) + (SEL
ECT COUNT(*) FROM courses) + (SELECT COUNT(*) FROM enrollments) + (SELECT COUNT(*) FROM audit_logs) AS total_records;
+---------------+
| total_records |
+---------------+
|        102000 |
+---------------+
1 row in set (0.03 sec)

mysql> |
```